

A partial positive answer for overcomplete sets in ordinal interval spaces

Agent lane 16

2026-06-23

Status. Candidate partial result, likely valid. This packet proves a ZFC positive subcase of Question 48 from Koszmider's arXiv:2006.00806, not the full $\xi < \omega_2$ problem.

1 Source question

The source paper is:

P. Koszmider, *On the existence of overcomplete sets in some classical nonseparable Banach spaces*, arXiv:2006.00806.

On page 25, Question 48 asks:

Can one prove in ZFC that the Banach spaces $C([0, \xi])$ for all ordinals $\xi < \omega_2$ admit overcomplete sets?

There are also some natural open questions for which concern more particular classes for Banach spaces like the following three questions:

Question 48. *Can one prove in ZFC that the Banach spaces $C([0, \xi])$ for all ordinals $\xi < \omega_2$ admit overcomplete sets?*

Partial results related to Question 48 are Theorem 11 (2) and result of 37 (Theorem 5 (3) in this paper). They cover all remaining ordinals as $C([0, \xi])$ admits a fundamental biorthogonal system $(1_{[0, \eta]}, \delta_\eta - \delta_{\eta+1})_{\eta < \xi}$ and $1_{[0, \xi]}, \delta_\xi$.

2 Definitions and source tool

Definition 1. *Let X be a Banach space. A set $D \subseteq X$ is overcomplete if $|D| = \text{dens}(X)$ and every subset $A \subseteq D$ with $|A| = |D|$ has dense linear span in X .*

We use the following criterion from the source paper, Theorem 10 there. For reference we state only the form needed here.

Lemma 1 (Koszmider's criterion). *Let X be a Banach space admitting a linearly dense family $\{x_\alpha : \alpha < \omega_1\}$. Suppose that there is a norm closed subspace $Y \subseteq X^*$ of finite codimension such that, for every $y^* \in Y$,*

$$\{\alpha < \omega_1 : y^*(x_\alpha) \neq 0\}$$

is countable. Then some finite-codimensional subspace of X admits an overcomplete set. More precisely, there is $k < \omega$ such that every codimension- k subspace of X admits an overcomplete set.

The source paper applies this with $k \leq 1$ to $C([0, \omega_1])$, and then uses the standard fact that, for the scattered compacta under consideration, finite-codimensional subspaces of $C(K)$ are isomorphic to $C(K)$. We use the same last step.

3 Idea of the proof

Koszmider's proof for $C([0, \omega_1])$ produces a dense family whose interaction with measures is countable after one endpoint mass is removed. For ordinals below $\omega_1 \cdot \omega$, the compact interval $[0, \xi]$ breaks into finitely many clopen ordinal blocks, each of type at most ω_1 . On each block the same countable-support criterion is available: countable blocks are separable, and ω_1 -blocks are exactly the source case. A finite ℓ_∞ -sum preserves the criterion because a finite union of countable exceptional sets is countable and the codimensions add only finitely.

4 The partial theorem

Theorem 1. *In ZFC, for every ordinal $\xi < \omega_1 \cdot \omega$, the Banach space $C([0, \xi])$ admits an overcomplete set.*

Proof. If $\xi < \omega_1$, then $C([0, \xi])$ is separable, and the existence of an overcomplete sequence is the classical theorem of Klee, quoted as Theorem 4 in the source paper. Thus assume $\omega_1 \leq \xi < \omega_1 \cdot \omega$.

Choose $m < \omega$ such that $\xi < \omega_1 \cdot m$. The ordinal interval $[0, \xi]$ is the finite union of clopen intervals

$$K_0, \dots, K_{m-1},$$

where empty final pieces are omitted, and each K_j is order-homeomorphic to an ordinal interval $[0, \eta_j]$ with $\eta_j \leq \omega_1$. Concretely, one cuts at the finitely many points $\omega_1, \omega_1 \cdot 2, \dots$ below ξ ; intervals of the form $(\gamma, \delta]$ in an ordinal compactum are clopen. Therefore

$$C([0, \xi]) \cong \bigoplus_{j < r}^{\ell_\infty} C(K_j)$$

for some finite r .

We first check that each summand $C(K_j)$ satisfies the hypothesis of Lemma 1, allowing a finite-codimensional subspace of its dual. If $\eta_j < \omega_1$, then $C(K_j)$ is separable; choose a countable linearly dense set D_j , and take $Y_j = C(K_j)^*$. For each $y^* \in Y_j$, the set of $d \in D_j$ with $y^*(d) \neq 0$ is countable simply because D_j is countable.

If $\eta_j = \omega_1$, then K_j is homeomorphic to $[0, \omega_1]$, and the source proof of Theorem 11(2) gives exactly the needed data: a linearly dense set $D_j \subseteq C(K_j)$ of cardinal ω_1 and a codimension-one subspace $Y_j \subseteq C(K_j)^*$ such that every $y^* \in Y_j$ is nonzero on at most countably many members of D_j .

Now pass to the finite ℓ_∞ -sum

$$X = \bigoplus_{j < r}^{\ell_\infty} C(K_j).$$

Embed each D_j into the j -th coordinate of X , and let

$$D = \bigcup_{j < r} \iota_j(D_j).$$

The set D is linearly dense in X . Since at least one block has density ω_1 in the present nonseparable case, $|D| = \omega_1$.

The dual is the finite ℓ_1 -sum

$$X^* = \bigoplus_{j < r}^{\ell_1} C(K_j)^*.$$

Set

$$Y = \bigoplus_{j < r}^{\ell_1} Y_j.$$

Then Y has finite codimension in X^* , since each Y_j has finite codimension and there are only finitely many summands. If $y^* = (y_j^*)_{j < r} \in Y$, then y^* is nonzero on the coordinate copy $\iota_j(D_j)$ only at the countably many points where y_j^* is nonzero on D_j . Hence y^* is nonzero on only a finite union of countable sets inside D , therefore on only countably many members of D .

Thus $X \cong C([0, \xi])$ satisfies Lemma 1. Consequently a finite-codimensional subspace of $C([0, \xi])$ admits an overcomplete set. Finally, $[0, \xi]$ is an infinite compact ordinal interval with a nontrivial convergent sequence. As used in the source proof of Theorem 11(2), this implies that finite-codimensional subspaces of $C([0, \xi])$ are isomorphic to $C([0, \xi])$. Since admitting an overcomplete set is preserved by Banach-space isomorphism, $C([0, \xi])$ itself admits an overcomplete set. $\square$

5 Limitations and novelty check

This does not solve Question 48 in full. The argument depends on having only finitely many ω_1 -sized ordinal blocks. It therefore reaches $\xi < \omega_1 \cdot \omega$, but it does not address ordinals such as $\omega_1 \cdot \omega$, ω_1^2 , or arbitrary $\xi < \omega_2$, where one would need either a countable-block version of Koszmider's criterion or a new selection argument.

Local run indexes were searched for arXiv:2006.00806, overcomplete-set keywords, $C([0, \xi])$, Grothendieck/scattered variants, and related ordinal phrases. The only close hit was a separate candidate full packet for arXiv:1912.08690 concerning weakly discrete overcomplete sets in $C([0, \omega_1])$; it does not cover the present ordinal-interval subcase. Web searches for the exact Question 48 wording and close variants did not find a later paper resolving this subcase or the full $\xi < \omega_2$ problem.

6 Human review recommendation

Review as a likely valid partial result. The main checks are:

1. the finite clopen decomposition of $[0, \xi]$ for $\xi < \omega_1 \cdot \omega$;
2. the finite-sum preservation of Lemma 1;

3. the final use of the source paper's finite-codimensional $C(K) \cong$ hyperplane fact for compact ordinal intervals with nontrivial convergent sequences.

References

- [1] P. Koszmider, *On the existence of overcomplete sets in some classical nonseparable Banach spaces*, arXiv:2006.00806.
- [2] T. Russo and J. Somaglia, *Overcomplete sets in non-separable Banach spaces*, arXiv:1912.08690.